

Christian DiPietrantonio
ME 5130: Advanced Heat and Mass Transfer
Homework 01: Heat Conduction
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1–2, Problem Statement: Verify that ∇T and $\nabla \cdot \mathbf{q}$ in the spherical coordinate system (r, ϕ, θ) are given as:

$$\nabla T = \hat{\mathbf{u}}_r \frac{\partial T}{\partial r} + \hat{\mathbf{u}}_\phi \frac{1}{r \sin \theta} \frac{\partial T}{\partial \phi} + \hat{\mathbf{u}}_\theta \frac{1}{r} \frac{\partial T}{\partial \theta} \quad (1)$$

$$\nabla \cdot \mathbf{q} = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 q_r) + \frac{1}{r \sin \theta} \frac{\partial q_\phi}{\partial \phi} + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (q_\theta \sin \theta) \quad (2)$$

1–2, Solution:

1–12, Problem Statement: Set up the mathematical formulation of the following heat conduction problems:

1. A slab in $0 \leq x \leq L$ is initially at a temperature $F(x)$. For times $t > 0$, the boundary at $x = 0$ is kept insulated and the boundary at $x = L$ dissipates heat by convection into a medium at zero temperature.
2. A semiinfinite region $0 \leq x < \infty$ is initially at a temperature $F(x)$. For times $t > 0$, heat is generated in the medium at a constant rate of $g_0 W/m^3$, while the boundary at $x = 0$ is kept at zero temperature.
3. A solid cylinder $0 \leq r \leq b$ is initially at a temperature $F(r)$. For times $t > 0$, heat is generated in the medium at a rate of $g(r)$, W/m^3 , while the boundary at $r = b$ dissipates heat by convection into a medium at zero temperature.
4. A solid sphere $0 \leq r \leq b$ is initially at temperature $F(r)$. For times $t > 0$, heat is generated in the medium at a rate of $g(r)$, W/m^3 , while the boundary at $r = b$ is kept at a uniform temperature T_0 .

1–12, Solution:

1–15, Problem Statement: A long cylindrical iron bar of diameter $D = 5$ cm, initially at temperature $T_0 = 650^\circ\text{C}$, is exposed to an air stream at $T_\infty = 50^\circ\text{C}$. The heat transfer coefficient between the air stream and the surface of the bar is $h = 80 \text{ W m}^{-2} \text{ }^\circ\text{C}^{-1}$. Thermophysical properties may be taken as $\rho = 7800 \text{ kg m}^{-3}$, $C_p = 460 \text{ J kg}^{-1} \text{ }^\circ\text{C}^{-1}$, and $k = 60 \text{ W m}^{-1} \text{ }^\circ\text{C}^{-1}$. Determine the time required for the temperature of the bar to reach 250°C by using the lumped system analysis.

1–15, Solution:

1–16, Problem Statement:

1–16, Solution:

2–2, Problem Statement:

2–2, Solution:

2–6, Problem Statement:

2–6, Solution:

2–15, Problem Statement:

2–15, Solution: